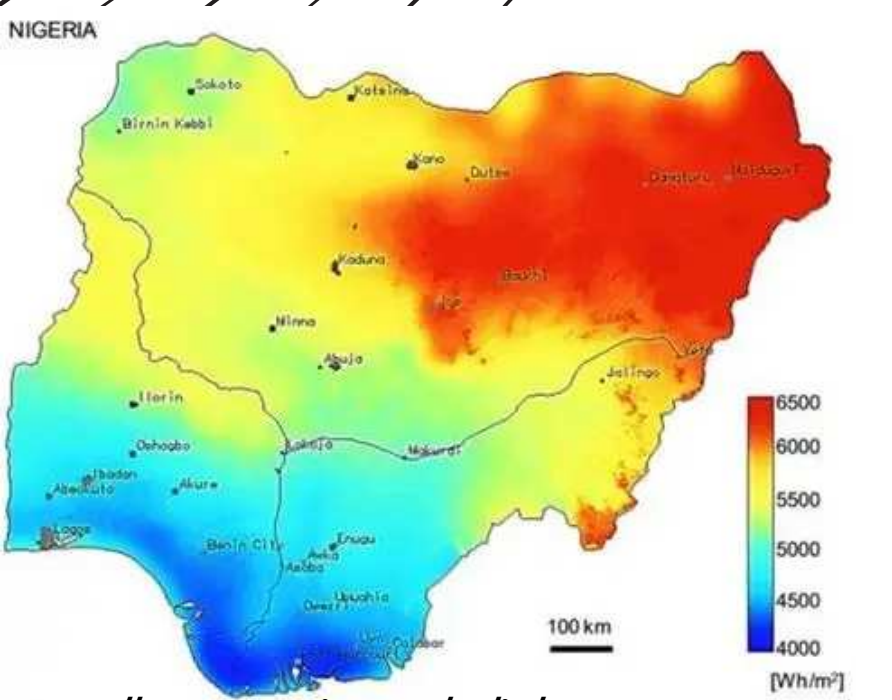


Background



- Northern regions of Nigeria have sustainable methods to utilize oil revenue
- Difficulty accessing oil due to high costs
- Challenges from Nigeria's government infrastructure

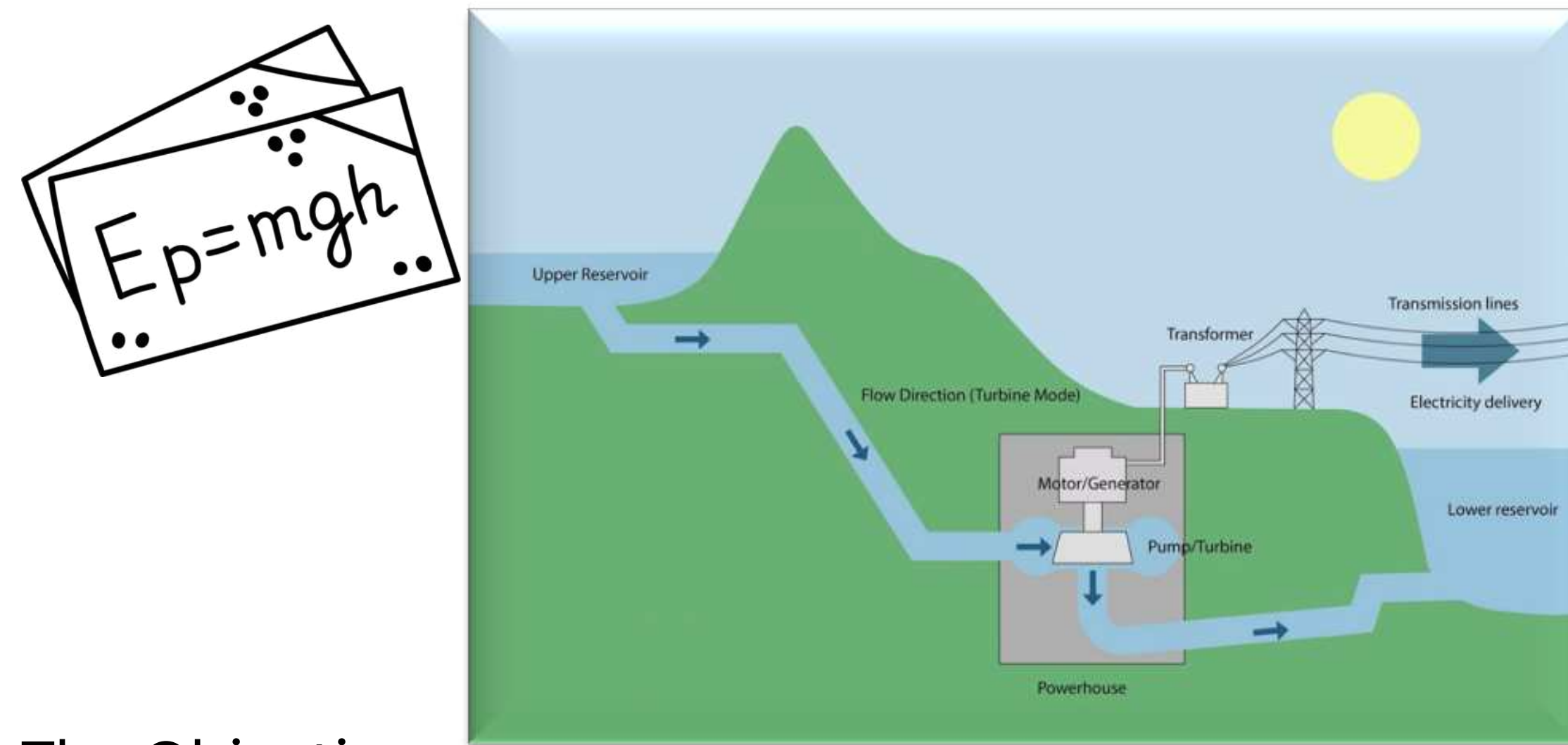
Because the northeastern region has a great source of solar radiation, it may be able to use it, although with the side effects of an unstable source of energy. In order to stabilize this energy, we use batteries as a buffer, where



These forms of energy storage use chemicals that, when they get damaged, would be exposed to the people, and cause kg of CO2 gases.

Solution

In order to work around the problems of storing energy and finding a stable replacement for chemical energy, we will be using a form of energy storage called gravitational energy storage, where energy is stored through potential energy.



The Objective:

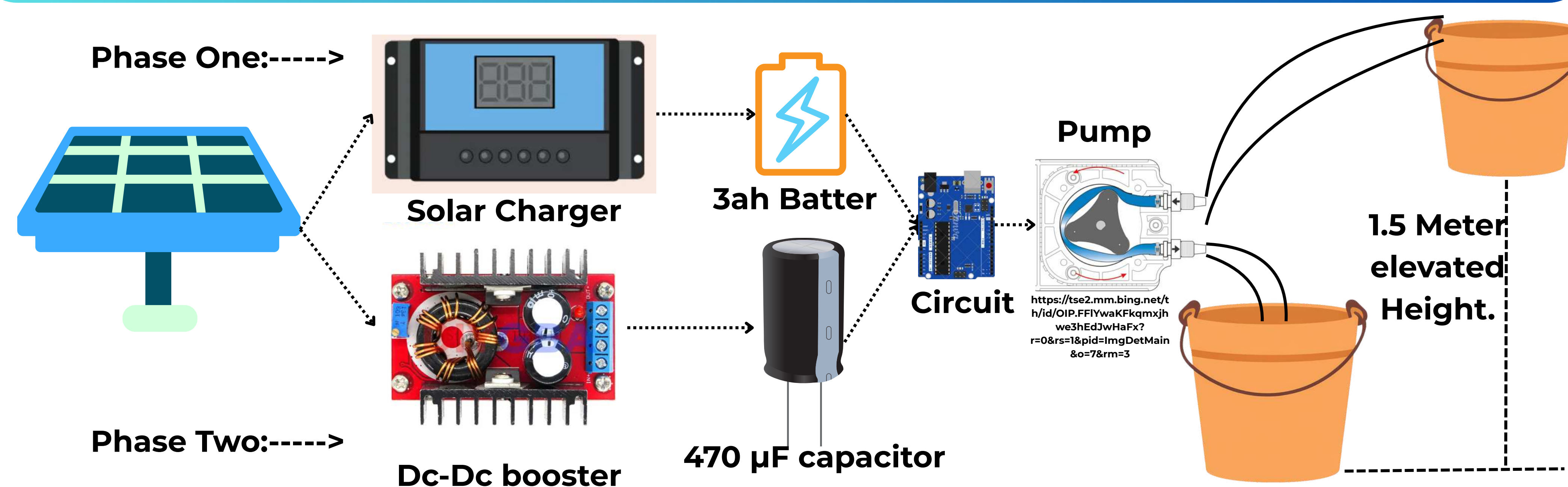
- Create an automated gravitational energy storage system
- Use a solar-powered pump for operation
- Store potential energy by lifting water
- Establish a direct connection from the solar panel to the motor
- No intermediary buffer needed between the solar source and the motor.

Hypothesis: By implementing a gravitational battery system, the approximately 50 % or more of the energy of the solar panel should be stored successfully.

Independent Variables (IV):-----> **Rate of water transferred**
Voltage (v)
Current (Amp)
Dependent variables (DV):---> **Energy Efficiency (%)**
Power (W)
Height (m)
Constants: -----> **Volume of Water used (liters)**

Mitigating solar intermittency by prototyping a low-resource, automated gravitational energy storage system designed to empower off-grid communities in Nigeria.

Methods



Phase 1:

Set up the solar charger's connection with the circuit and battery, using a 100 µF capacitor.

Phase 2:

Set up the DC-DC booster and connect it to the 470 capacitor (replacing the 100 µF).

Data collection process: (Set up phase 1 first, then phase 2):

- Set up two buckets, with one being elevated, the other filled with water. Power the Circuit (Raspberry Pi powers the Arduino) and set the code ready to start in the terminal (Raspberry Pi takes the data from the Arduino: the voltage and current, and puts it in a CSV file).
- Power is applied from the solar panel through phase 1 or 2 to the motor circuit. The Arduino commands the A4988 (controls the NEMA stepper motor) to begin turning. For phase 2, it will slowly accelerate to the motor's maximum speed for torque.
- Start the command to let the Raspberry Pi collect data from the Arduino. Let it run till the water sensor senses water, and it stops on its own. Measure mass every 30 mins

```
import serial
import time
import sys

connect to Arduino
ser = serial.Serial('/dev/ttyUSB0', 115200, timeout=1)

print("Research Mode: Recording data until water is detected...")

with open("pump_research_data.csv", "w") as file:
    file.write("Timestamp, Volts, Amps\n")

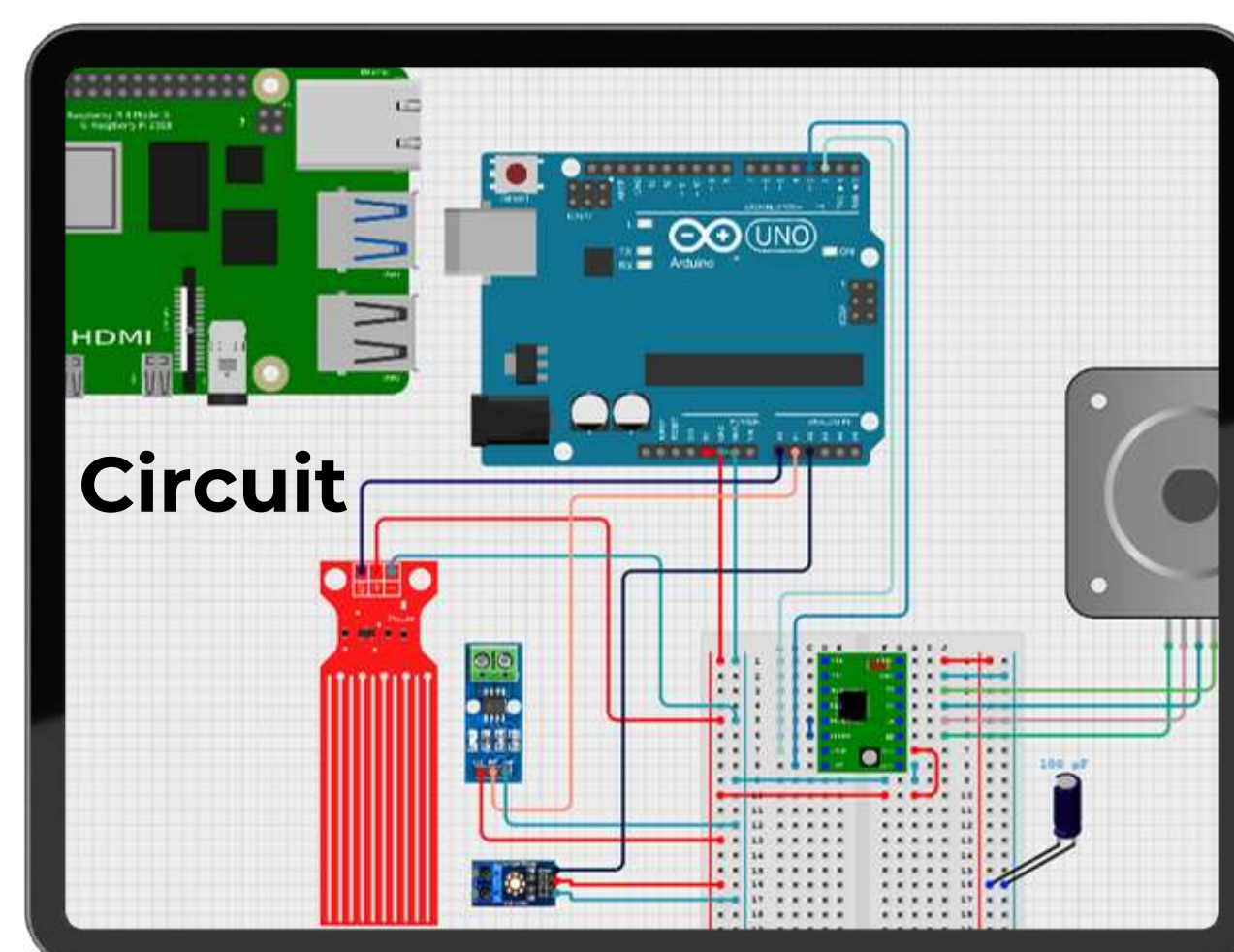
    try:
        while True:
            if ser.in_waiting > 0:
                line = ser.readline().decode('utf-8').strip()

                # Check if the Arduino sent the Stop signal
                if line == "STOP":
                    print("Data Collection complete. File saved.")
                    break # This exits the while loop

            # Otherwise, process the numbers
            try:
                volts, amps = line.split(",")
                now = time.strftime("%H:%M:%S")
                print(f"[{now}] Battery: {volts}V | Current: {amps}A")
                file.write(f"{now}, {volts}, {amps}\n")
                file.flush()
            except ValueError:
                continue

    except KeyboardInterrupt:
        print("Manual stop detected.")

# The script reaches here after the 'break'
ser.close()
sys.exit()
```



Example data

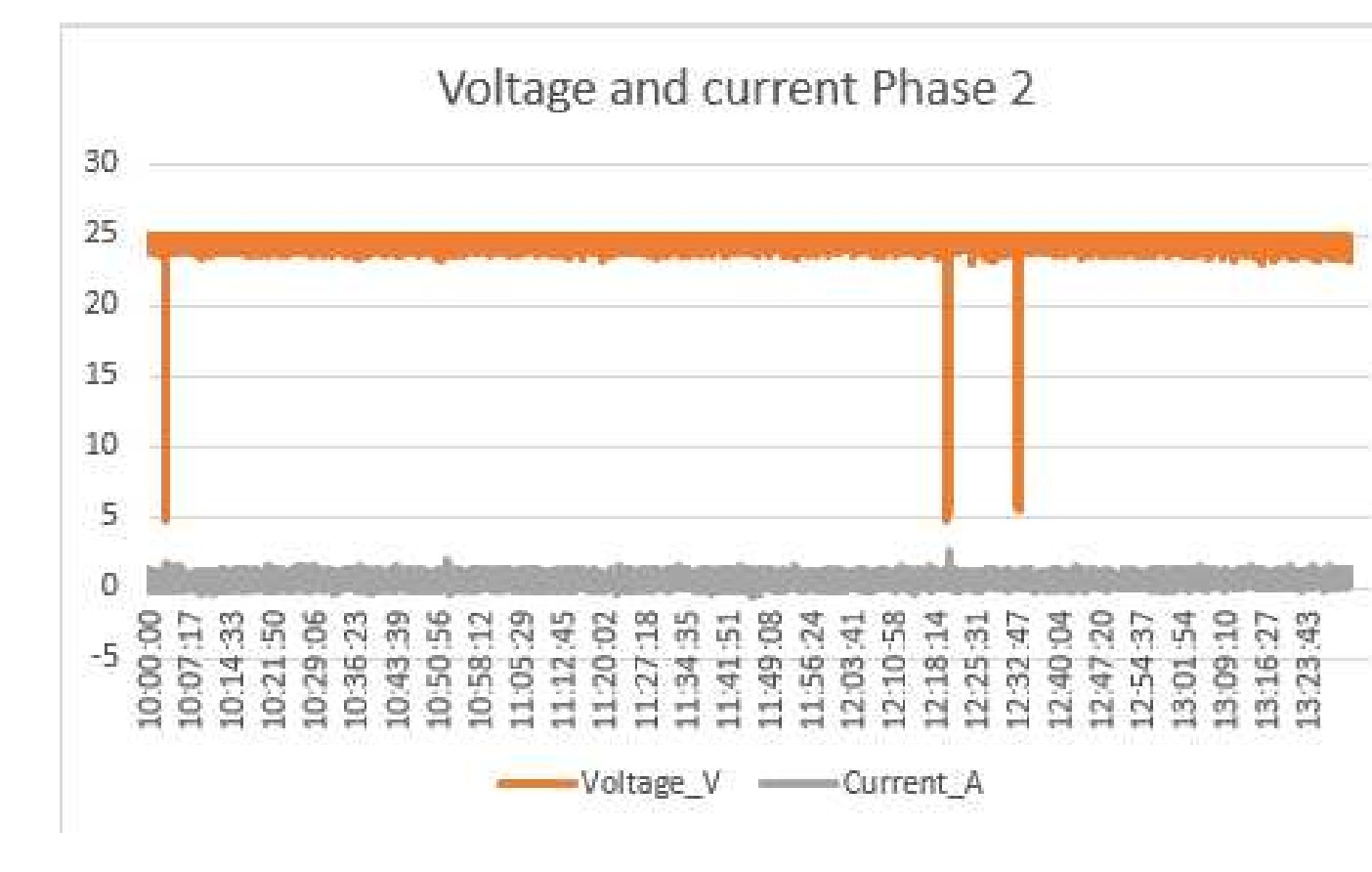
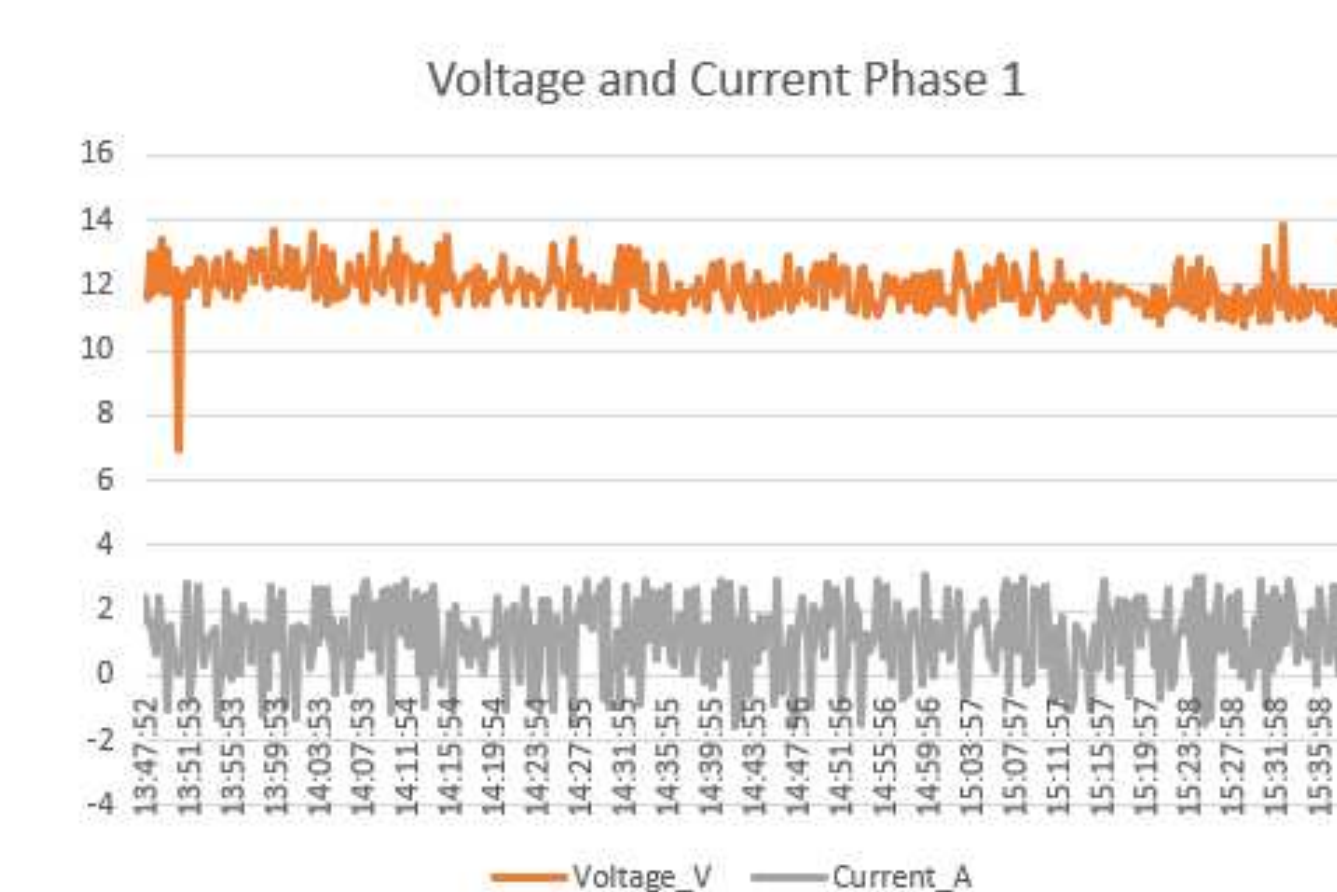
Timestamp	Voltage_V	Current_A	Power_W
13:47:52	11.82	1.37	16.1934
13:48:07	11.77	2.44	28.7188
13:48:22	11.57	1.61	18.6277
13:48:37	12.99	1.71	22.2129
13:48:52	11.74	1.07	12.5618
13:49:07	11.87	0.68	8.0716
13:49:22	12.87	2.39	30.7593
13:49:37	13.4	1.22	16.348
13:49:52	11.79	1.46	17.2134
13:50:07	13.09	-1.03	-13.4827
13:50:22	11.77	1.51	17.7727
13:50:37	11.84	0.78	9.2352
13:50:52	12.5	0.29	3.625
13:51:07	6.93	0.05	0.3465
13:51:22	12.26	0.39	4.7814
13:51:38	12.18	1.22	14.8596
13:51:53	11.67	2.83	33.0261
13:52:08	12.43	-0.73	-9.0739
13:52:23	12.45	0.24	2.988

Data

Time interval	average power every 30 min	Energy Collected	Water Transferred (m)	Water Transferred (L)	Energy Stored	Energy efficiency
10:00-10:30	13.11085178	23599.5332	180.6	0.1806	2.65482	1.12%
10:30-11:00	13.00097881	23401.76186	270.9	0.2709	3.98223	1.70%
11:00-11:38	12.74399335	22939.1883	180.6	0.1806	2.65482	1.16%
11:38-12:00	12.64641608	22763.54895	120.4	0.1204	1.76988	0.78%
12:00-12:30	12.82152297	23078.74135	180.6	0.1806	2.65482	1.15%
12:30-1:00	14.01207458	25221.73425	150.5	0.1505	2.21235	0.88%
1:00-1:30	14.67415548	26413.47987	120.4	0.1204	1.76988	0.67%
Total:		167417.9878	1204	1.204	17.6988	1.06%
Direct_test						
L/Hour:		344	0.344			
power used to transfer water:		0.038625412	3.86254E-05			

Time interval	average power every 30 min	Time Difference (hour)	Energy Collected	Water Transferred (m)	Water Transferred (L)	Energy Stored	Energy efficiency
1:48-2:14	14.37086	1620	23280.7932	850	0.85	12.495	5.37%
2:14-2:30	13.15511429	960	12628.90971	950	0.95	13.965	11.06%
2:30-3:03	13.55664848	1980	26842.164	2290	2.29	33.663	12.54%
3:03-3:25	13.88216818	1320	18324.462	1790	1.79	26.313	14.36%
3:25-3:39	14.00031754	840	11760.26674	450	0.45	6.615	5.62%
Total:			92836.59565	6330	6.33	93.051	10.02%
L/Hour:			1808.571429	1.808571429			

Battery test



Results

- When comparing the direct test to the battery-assisted storage test, a noticeable difference in data is observed.
- The amount of water transferred is significantly lower in the direct test than in the battery-assisted test.
- This difference is also reflected in the energy efficiency results.
- The average efficiency for the direct test was approximately 1.06%.
- In contrast, the battery-assisted test achieved an average efficiency of around 10.02%.
- This means that the efficiency of the battery-assisted test is ten times greater than that of the direct test.
- Overall, these findings highlight the advantages of battery assistance in improving both water transfer and energy efficiency.

Energy effecncy	Energy effecncy
1.12%	5.37%
1.70%	11.06%
1.16%	12.54%
0.78%	14.36%
1.15%	5.62%
0.88%	10.02%
0.67%	
1.06%	

Battery

L/Hour :	1808.571429	1.808571429
power used to tranfer water:	0.007626473	0.00000762647

Direct

L/Hour :	344	0.344
power used to tranfer water:	0.038625412	0.0000386254

Discussion/Conclusion

- The initial hypothesis of the research was not fully supported by the prototype.
- The prototype achieved a maximum energy efficiency of only 14.36%.
- Data collected offers crucial insights into low-resource mechanical systems for energy storage.
- Peristaltic pumps require consistently high starting torques to maintain compression during motion.
- Direct solar energy is affected by passing clouds and changing sun angles.
- Changes in sunlight can cause the motor to stall, leading to wasted energy as heat instead of being converted into kinetic movement.

To address these issues, further research should explore replacing the peristaltic pump with alternative pump designs and investigate other variables, such as height and mass. Due to time constraints, not all variables were able to be tested simultaneously. However, with additional time and resources, this research has the potential to achieve energy efficiencies of 50% or more, while maintaining the low-resource automated gravitational system that mitigates the effects of solar fluctuations.

References

Bowoto, Oluwole K., et al. 'Gravitricity Based on Solar and Gravity Energy Storage for Residential Applications'. International Journal of Energy and Environmental Engineering, vol. 12, no. 3, Sept. 2021, pp. 503-16. Springer Link, Your paragraph text. Leccese, F., Cagnetti, M., Di Pasquale, S., Giarnetti, S., & Caciotta, M. (2016). A new power-quality instrument based on Raspberry Pi. Electronics, 5(4), 64. Your paragraph text Ruoso, A. C., Caetano, N. R., & Rocha, L. A. O. (2019). Storage gravitational energy for small-scale industrial and residential applications. Inventions, 4(4), 64. Your paragraph text Google. (2026). Gemini (Gemini 3.1 Pro) [Large language model]. <https://gemini.google.com>